

FastReact – A Latency Resilient Distributed Quiz System

Github Repo: <https://github.com/wymenlim/IOT-Project>
BSc (Hons) in Computing Science

Introduction and Problem Statements

Problem Statement: How can a distributed multi-hop wireless IoT system guarantee fair, low latency, secure, and scalable first responder determination independent of network congestion, routing depth, and packet arrival timing?

Fairness in determining a first responder is critical, yet traditional systems rely on unreliable packet arrival times. Network jitter, congestion, and varying MAC-layer mechanisms introduce delays, ensuring that the first packet received is not always the first button pressed. This project utilizes ESP-NOW and AODV-like routing to build a distributed multi-hop network. By employing local timestamp-based arbitration, each node measures reaction time relative to a synchronized "GO" signal. This decouples fairness from network latency and hop depth. The system integrates multi-hop scalability with lightweight security, ensuring deterministic, fair results in dynamic, congestion-prone IoT environments.

Methodology and Proposed Solutions

Methodology & Flow

1. The server initiates the network by broadcasting an AUTH_REQ beacon. Players use this to learn the Server's MAC address and initiate their own Route Requests (RREQ) to establish a fresh path
2. Nodes send periodic RREQs (boot/manual/keepalives) to ensure the mesh routing table stays updated even if devices move. The server auto-registers unknown player MACs upon receiving these requests.
3. Active Gameplay:
 - a. Start Trigger: Server broadcasts a GO packet to all routed players.
 - b. Players capture the local timestamp immediately upon receipt to start the reaction timer.
 - c. When button is pressed, the player sends a PRESS packet containing the reaction time (in milliseconds) back to the server.
4. The server validates the press and sends an ACK. If a player misses an ACK, they automatically retry the transmission or re-run route discovery to find a new path to the server.
5. Determining the winner: Once all presses are received or a timeout occurs, the server computes the fastest time (including ties) and sends RESULT packets for UI updates.

Proposed Solutions

- **Unicast Security:** Implement ESP-NOW Link-Layer Encryption so that every unicast peer (Server to Node) uses a Primary Master Key (PMK) and a Local Master Key (LMK) to prevent eavesdropping on private game data.
- **End-to-End Authentication:** Every packet is signed with an HMAC-SHA256 signature. This ensures that even if a packet is relayed through multiple "untrusted" nodes, the final receiver can verify it hasn't been tampered with.
- **Immutable Authentication Fields:** Routing fields (like hop_count and ttl) must change as a packet moves, which usually breaks digital signatures. To combat this, the authentication hash is calculated only over "stable" fields. Relays can update routing metadata (sender MAC/TTL) without invalidating the end-to-end signature.
- **Replay & Duplicates:** A "Seen Table" tracks the origin_mac combined with a unique packet_id. By applying a short expiry time to these entries, the system prevents "Replay Attacks" (where an attacker resends an old "fast" press packet to win).
- **Protocol Version Check:** Each packet includes a protocol version check. This prevents "Version Rollback" attacks and ensures that all M5Stick nodes are operating on the same logic and timing rules, maintaining competitive integrity.

Research Questions and Project Objectives

The primary objective of this project is to design, develop, and evaluate a distributed multi-hop wireless button system capable of millisecond-accurate and fair winner determination under dynamic network conditions.

Research Questions

How to ensure the actual fastest player wins, considering all the latency for communication? How does the system maintain integrity even when there is congestion, relay failures, and conflicts?

Experiments and Results Analysis

Experiment 1: Packet Delivery Reliability (Direct, No Relay)

- **Setup:** Two player nodes within direct ESP-NOW range of the server.
- **Results:** 100% packet send success rate across all rounds. Only 1 ACK timeout occurred, which auto-recovered via retry without interrupting the round. For the 246 replay drops and 65 auth failures: Seen Table correctly rejects duplicate route discovery floods.
- **Analysis:** ESP-NOW's connectionless delivery proved highly reliable in direct topology. The retry mechanism functioned correctly as a fallback, confirming the reliability layer works as designed.

Experiment 2: Reaction Time Fairness

- **Setup:** Two players competed across 19 rounds. Winner correctness verified against raw reaction_ms in server logs.
- **Results:** The server correctly identified the winner in all rounds with zero fairness errors. Mean spread was 574.6ms, with tight rounds under 200ms. Outlier rounds showed spreads above 2400ms due to one player pressing late, not a system fault.
- **Analysis:** Local timestamp arbitration successfully decoupled fairness from network conditions. Since reaction_ms is measured on the player node at button press, network jitter had no impact on winner determination.

Experiment 3: Multi-Hop Relay Performance

- **Setup:** One player node placed out of direct range of the server, forced to route through an intermediate relay node. 10 rounds conducted. Hop count verified via serial logs.
- **Results:** Packet send success rate: 100%. Mean round-trip latency: 3844 ms (min 33 ms, max 12850 ms). ACK timeouts: 2, both auto-recovered via retry. All rounds completed within the 15-second timeout. Hop count confirmed at 1 for the relayed player.
- **Analysis:** Packet delivery remained perfect across all 10 relay rounds, consistent with direct topology results. The higher mean latency reflects slower human reaction times during this session rather than relay overhead. The relay hop itself adds only milliseconds of forwarding delay. Zero fairness errors were recorded, confirming that local timestamp arbitration remains accurate regardless of hop depth.

Metric	Result (Direct)	Result (Relay)
Delivery Success	100.0%	100%
Mean Latency	409.3 ms (player reaction time)	3844 ms (player reaction time)
Throughput	22.2 pkt/s (~2.8% Capacity)	3.8 pkt/s (~0.5% capacity)
Filtered Threats	311 packets (Auth/Replay)	0 packets
Winner Accuracy	100% Accurate	100% Accurate

Conclusions and Future Work

All in all, FastReact demonstrates that millisecond-fair, secure, and relay-resilient winner determination is achievable over a multi-hop ESP-NOW mesh by decoupling fairness from network latency through local timestamp arbitration.

Future Work: 1. Investigate dynamic channel selection to improve reliability in noisy RF environments; 2. Introduce server failover or leader election to eliminate single-point failure risk; 3. Transition reaction timer from milliseconds to microseconds via esp_timer_get_time() for finer press resolution; 4. Explore hybrid architectures or alternative protocols to scale beyond ESP-NOW's 20-peer encrypted peer limit.

References and Acknowledgement